

ARTHDRISHTI

FINANCIAL INTELLIGENCE FOR BHARAT

"Your bank sees transactions.
ArthDrishti sees financial life."

Team **pizza** · an AI-powered Financial Intelligence Layer for banking

OBSERVE → UNDERSTAND → PREDICT → **GOVERN** → HELP

THE PROBLEM

India's digital rails — UPI, net banking, video KYC — are world-class. The experience built on them is not. Customers in Tier 2/3/4 towns find banking apps confusing, generic, and disconnected from their real financial needs — even though banks already hold rich transactional and behavioral data.

- REQ-01Offers arrive as generic pop-ups, not at the right moment for the right customer.
- REQ-02First-time digital users get no vernacular, conversational path through onboarding, KYC, or loans.
- REQ-03Stress and fraud signals surface late — and are handled punitively, not empathetically.

THE INSIGHT

Personalization should be state-aware, not profile-only.

A profile is static; a financial life is not. ArthDrishti builds a continuously updated **financial state** per customer and decides what is appropriate *now*.

RAVI32 · salaried · digital banking user — representative, illustrative scenario

His bank sees:

salary creditsUPI transactionsEMIs

spending categories

savings

"Mujhe ₹5 lakh ka personal loan lena chahiye?"

A CONVENTIONAL APP SEES
LOAN OPPORTUNITY

vs

ARTHDRISHTI SEES
FINANCIAL CONTEXT

Cash flow↓EMI burden↑Savings buffer↓Repayment stress↑

→

GOVERNANCE CORE

NO LOAN RECOMMENDATION→

INSTEAD

FINANCIAL GUIDANCE

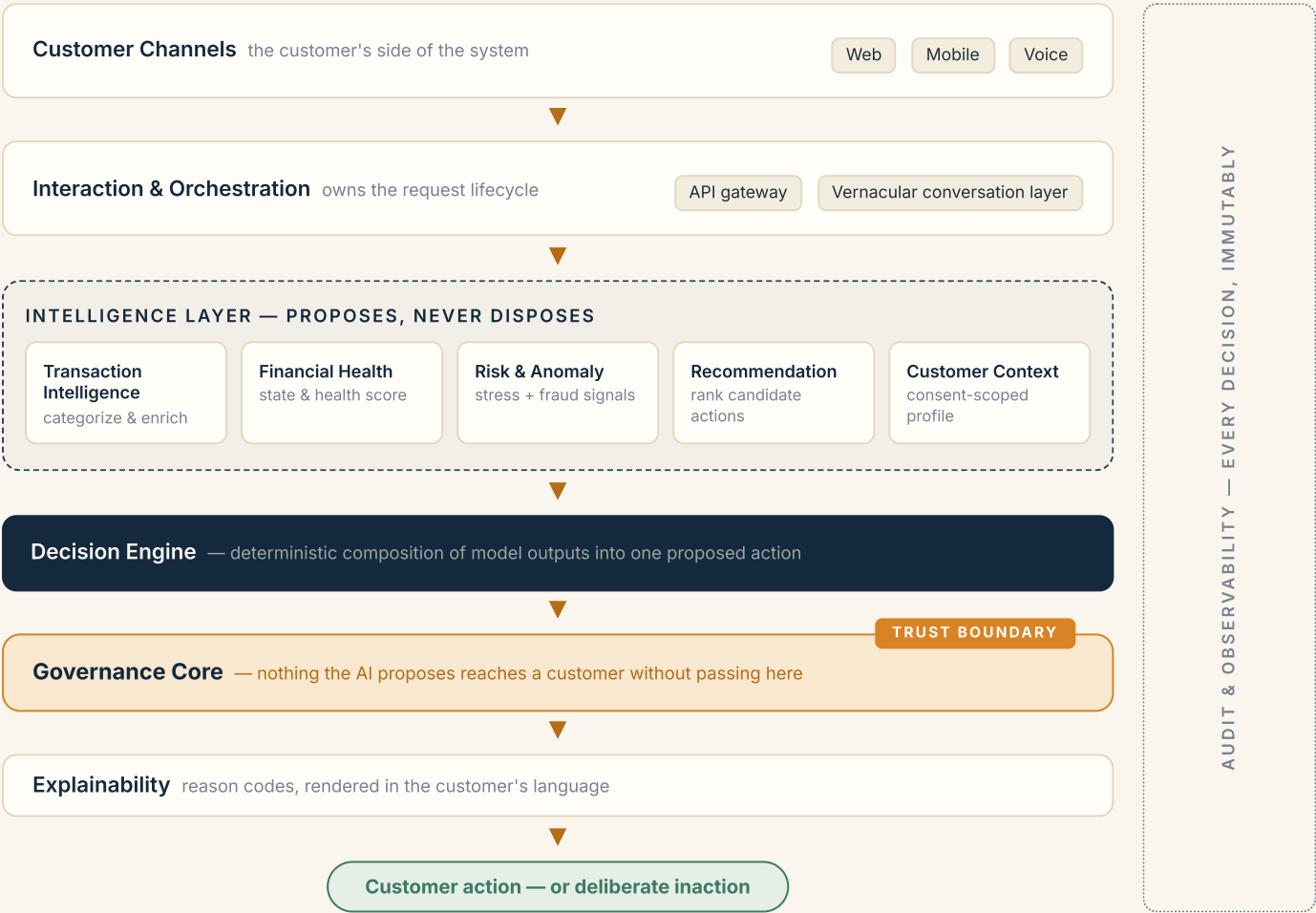
His bank does not see how his situation is *changing* — ArthDrishti does. Sometimes the smartest financial recommendation is **no**.

CONVENTIONAL BANKING ASKS
"What can we sell?"

ARTHDRISHTI ASKS
"What does this customer actually need?"

From transactions to financial intelligence.

The architecture separates interaction, intelligence, decisioning, and governance. Intelligence services *propose*, the Decision Engine *composes*, and the Governance Core *disposes*. ARCH-01 · D-02



TECHNOLOGY — PROPOSED, NOT MANDATED

FRONTEND React / Next.js	BACKEND Python / FastAPI	INTELLIGENCE scikit-learn / XGBoost	LANGUAGE Multilingual LLM / Indic NLP
DATA PostgreSQL / pgvector	REAL-TIME Redis / Kafka	DEPLOYMENT Docker · cloud, India region	OBSERVABILITY Prometheus / Grafana

Interpretable tabular models where money is decided; an LLM only where language is the problem; one database with vector search rather than an extra store; India-region deployment to support data-localization design intent. The problem statement marks technology as advisory.

- 1 **Intelligence before interface** — the conversation renders decisions; it doesn't make them.

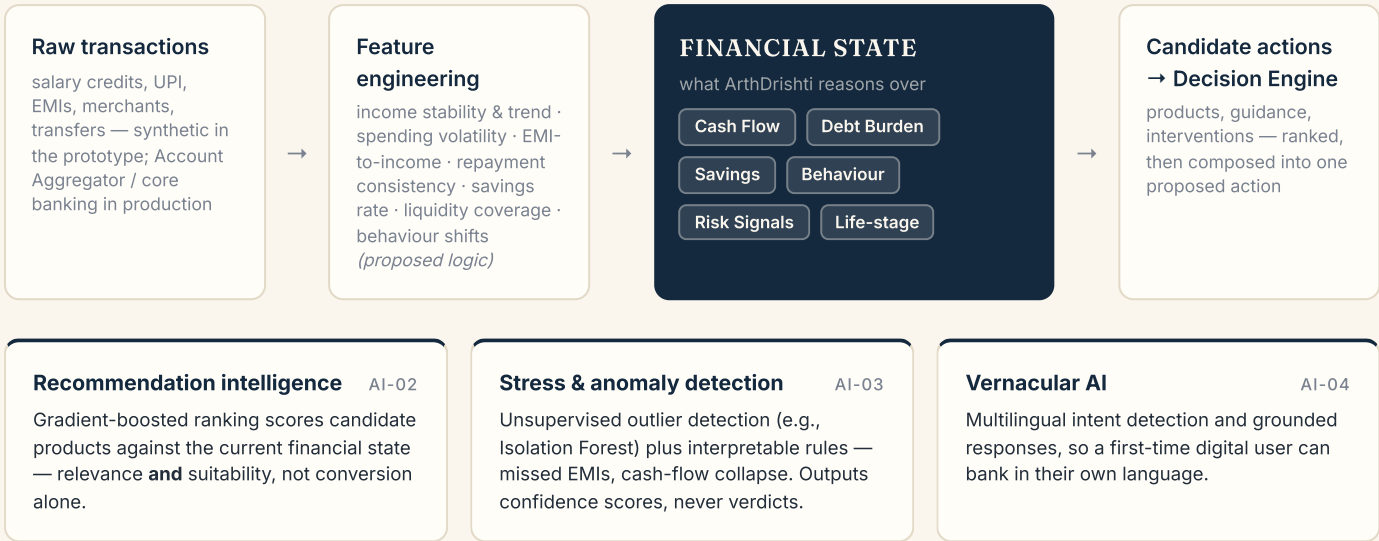
3 **Explainability by design** — every decision carries reason codes at creation.

5 **Least-privilege, consent-scoped access** — services see only purpose-approved data.
- 2 **Governance before action** — no customer-visible action bypasses the Governance Core.

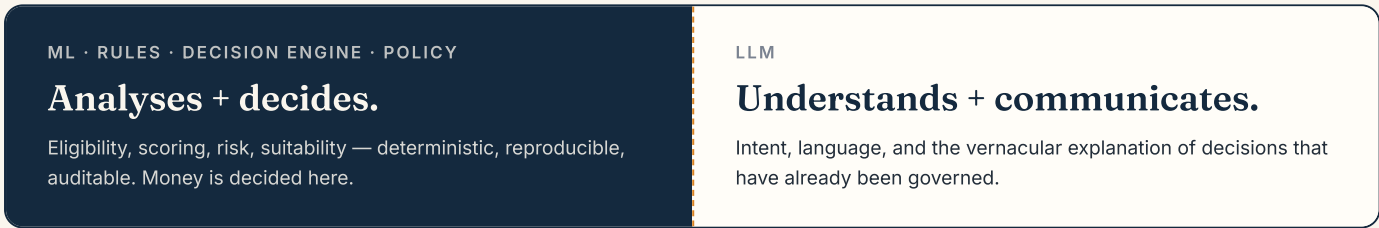
4 **Customer benefit over upsell** — suitability can veto revenue.

A financial state. Not just a profile.

Raw transactions are never mapped directly to products. They become a structured, versioned **financial state** — the single object all downstream decisioning consumes. AI-01 · D-03

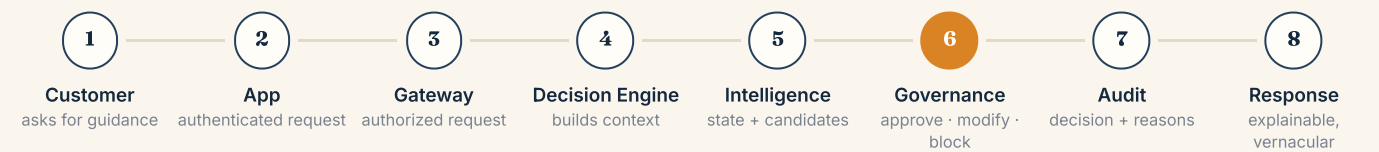


WHERE AI STOPS AND GOVERNANCE BEGINS



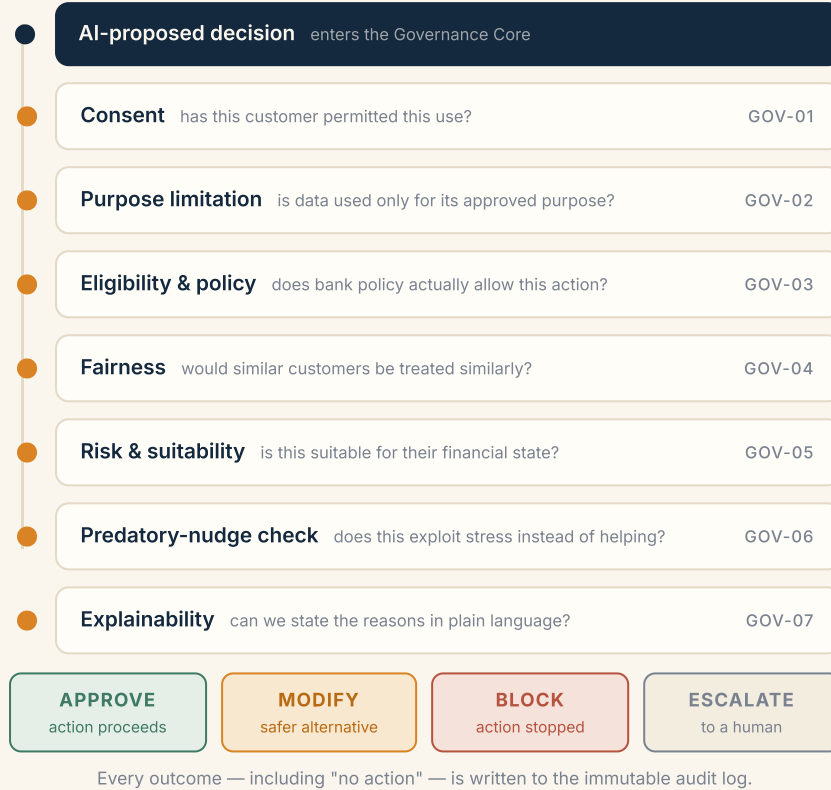
AI-05 The LLM never approves products, determines eligibility, calculates risk, or overrides governance. **Never.**

ONE CANONICAL REQUEST LIFECYCLE — every journey (dashboard, chat, proactive nudge) is an instance of it · D-04



Trust is not a footnote. It is a control plane.

The problem statement requires safeguards for data privacy and consent (DPDP Act, RBI data localization norms), algorithmic bias, and predatory nudging. ArthDrishti implements them as an architectural gate every AI-proposed decision must pass — not a checklist. D-05



ARTHDRISHTI CAN SAY NO.

Personalization is not about maximizing product conversion. It is about maximizing customer benefit. **"No product" is a valid decision.**

Financial stress ↑ + debt burden ↑ +
savings buffer ↓
→ loan suitability ↓
→ **GOVERNANCE CORE**
→ **no product**
→ financial guidance + repayment
support

"A financially stressed customer is not a sales opportunity. They are a customer who needs help."

SECURITY & PRIVACY · SEC-01 · D-06



Encryption in transit and at rest · tokenized customer identifiers inside AI services · consent recorded per purpose, revocation honored downstream · India-region data residency as a deployment constraint.

Access: RBAC sets the ceiling, ABAC decides each request

Roles (customer, relationship manager, risk officer, auditor, admin) grant a maximum; attributes (consent scope, purpose, data sensitivity) decide every individual request.

Lifecycles, state-machined

Role, request, and recommendation lifecycles are explicit state machines — provisioned → verified → active → suspended/deactivated — with every transition audited. Full detail in the technical appendix.

Regulatory posture: designed for regulatory readiness and alignment with DPDP and RBI expectations — design alignment, not legal compliance certification.

See ArthDrishti think.

Illustrative prototype interaction on synthetic data — not real financial advice. Values shown are illustrative.

CUSTOMER

"Mujhe ₹5 lakh ka personal loan lena chahiye?"

ARTHDRISHTI READS THE FINANCIAL STATE — before it answers

EMI / income		41% · high
Savings buffer		0.8 months · low
Cash flow		declining
Repayment behaviour		2 recent EMI delays
Risk signals		elevated stress

GOVERNANCE CORE LOAN OFFER BLOCKED

GOV-06 · predatory-nudge check
a stressed customer is not a sales opportunity

ARTHDRISHTI

"Aapke haaliya cash-flow aur maujooda EMI bojh ko dekhte hue, naya loan financial stress badha sakta hai — isliye abhi loan recommend nahi kar rahe."

Given your recent cash flow and current EMI burden, another loan may increase financial stress — so we are not recommending one now.

WHY?

High EMI burden · low savings buffer · recent repayment stress — every reason inspectable by the customer.

INSTEAD

Repayment guidance + cash-flow improvement suggestions — help first, products later.

WHAT WE BUILD FOR THE HACKATHON

- Synthetic transaction data + categorization
- Financial-state engine
- Recommendation engine
- Stress / anomaly detection
- Vernacular conversational layer
- Governance Core
- Explainable dashboard, end-to-end journey

Later: event-driven updates, scenario simulation, voice, model monitoring.

Production path (not the prototype): core-banking / Account Aggregator integration, production identity, certification.

MEASURING SUCCESS — targets; no measured results yet

Recommendation

relevance / Precision@K

Anomaly detection

false-positive rate

Explainability

explanation coverage

System

p95 latency / error rate

Stress detection

precision / recall / F1

Intent detection

accuracy per language

Governance

unsafe-action block rate

Fairness

group disparity metrics

PATH TO SCALE

- Synthetic data
- ↓ Account Aggregator / core banking
- ↓ Event-driven financial state
- ↓ Independently scalable AI services
- ↓ Monitoring + model governance

The prototype's service boundaries are the production boundaries — hardened, not rewritten.

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Team **pizza** · HackOut'26 · complete engineering artifacts — HLD/LLD, DFDs, ER & schemas, API/event contracts, state machines, threat & privacy models, CI/CD — live in the project repository.

Understand before recommending.

Govern before acting.

Explain before influencing.

Protect before selling.